SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – FLOWCHART-TO-CODE CONVERSION

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	**Assessment:**	Assessment Tool 2 – Flowchart-To-Code Conversion
Weightage:	35% of CIA	**Total Marks:**	100
CO5 Statement:	*Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.*		
Student Name:	T. Varun Tej	**Reg. No.:**	192571217
Section / Batch:		**Date:**	04/09/2026

Code of Conduct

I T. Varun Tej / 192571217 *(Name / Reg No)*

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Varun

Instructions

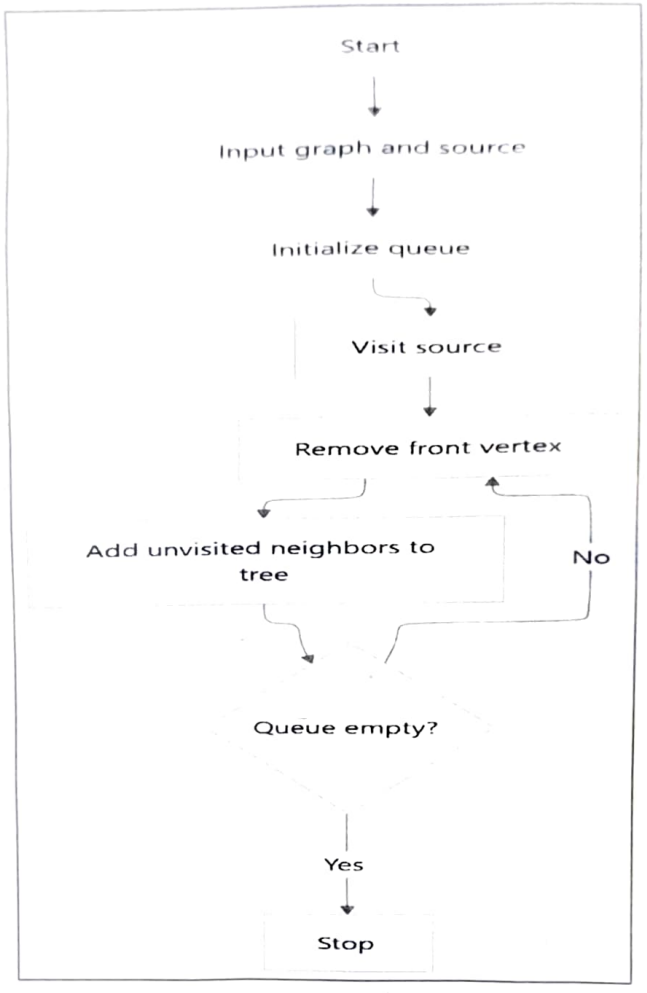
- This test contains 5 Flowchart-to-code conversion questions. Total: 100 Marks.
- When a student answer using Flowchart-to-code conversion should evaluate based on the ability to design clear, logical, and efficient code solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Graph Traversal	BFS	1	20
Shortest Path Algorithm	Dijkstra's Algorithm	2	20
Shortest Path Algorithm	Unweighted Graph	3	20
Graph Traversal	BFS	4	20
Minimum Spanning Tree	Prim's Algorithm	5	20
	GRAND TOTAL:		**100 Marks**

Annexure A – Pseudocode Writing Task

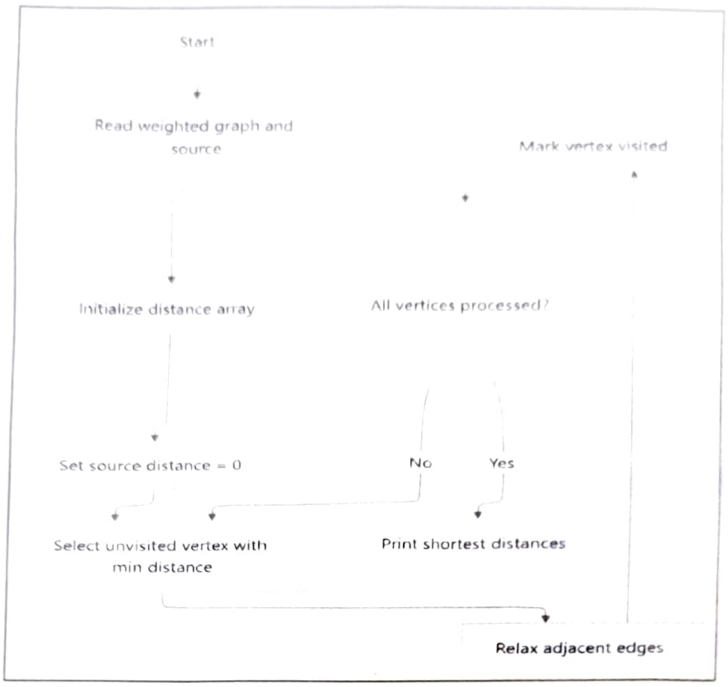
1. Convert the flowchart for generating a BFS tree into code. (20 Marks)

Flowchart Diagram:



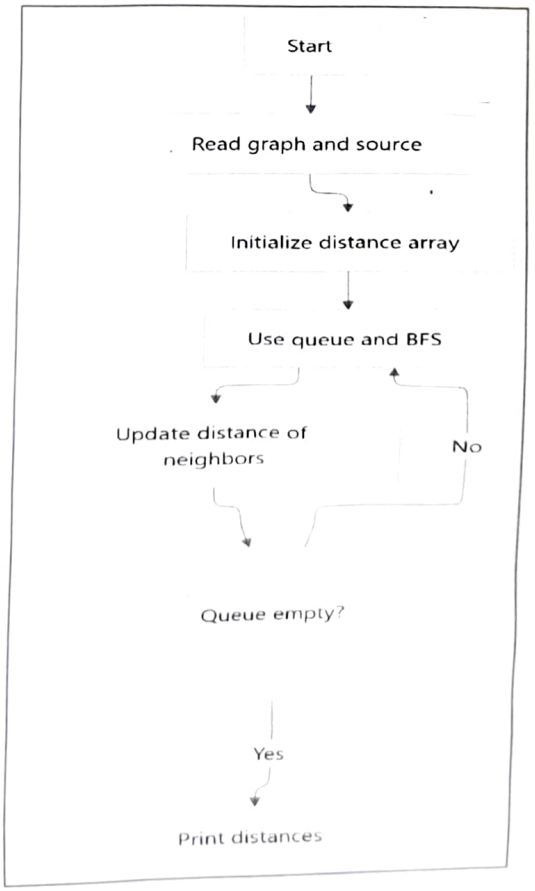
2. Convert the flowchart for Dijkstra's shortest path algorithm into code. (20 Marks)

Flowchart Diagram:



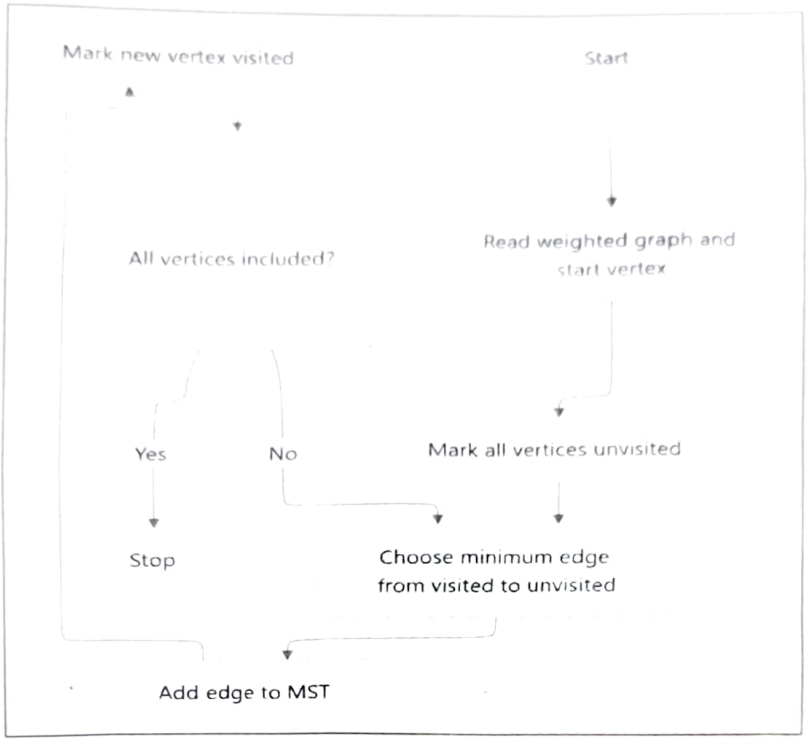
3. Convert the flowchart for shortest path in an unweighted graph into code. **(20 Marks)**

Flowchart Diagram:



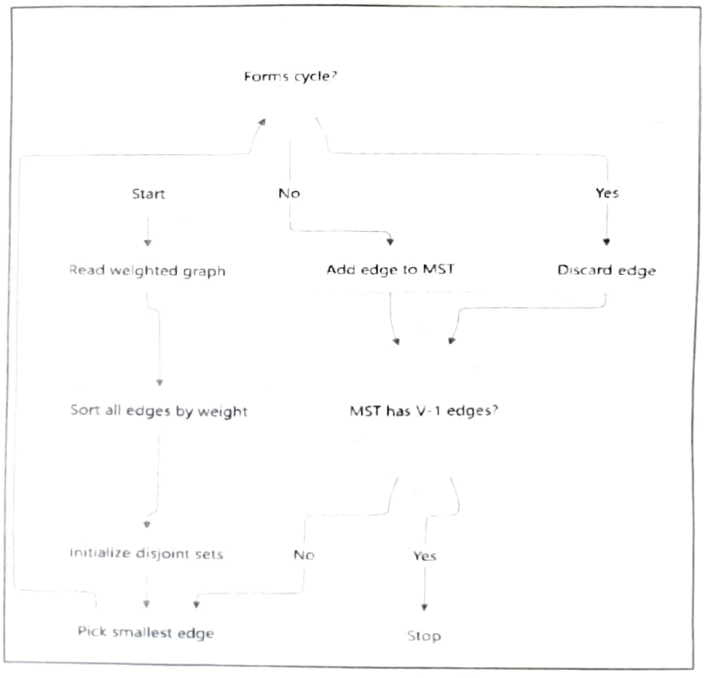
4. Convert the flowchart for Prim's algorithm into code. (20 Marks)

Flowchart Diagram:



5. Convert the flowchart for Kruskal's algorithm into code. (20 Marks)

Flowchart Diagram:



Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding, handles edge cases effectively	Good understanding, minor gaps in edge case handling	Partial understanding, misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct, minor errors	Partially correct; several test cases fail	Incorrect or non-functional code
Code Quality & Readability	20	Clean, modular, well-commented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	Course Coordinator	Program Director
Signature:	Signature:	Signature:
Date:	Date:	Date:

CO5 - AT-2

Name : T Varun Teja
Reg No : 192511217
Code : CSA0305
Course : Data Structures

1) Convert the flowchart for generating a BFS tree into code

A)

```
def bfs-tree(graph, source):
    visited = set([source])
    queue = deque([source])
    bfs-tree-edges = []
    transversal-order = []
    while queue:
        u = queue.popleft()
        transversal-order.append(u)
        for v in graph.get(u, []):
            if v in not in visted:
                visited.add(v)
                queue.append(v)
                bfs-tree-edges.append([u,v])
    return transversal-order-bfs-tree-edges.
```

2) Convert the flowchart for Dijistra's shortest Path algorithm into code:

Ans:

```
import heap:
    def dijistra(graph, source):
        dist = {node: float('inf') for node in graph}
```

```
    dist[source] = 0
    visited = set()
    pq = [(0, source)]
while pq:
    d, u = heapq.heappop(pq)
    if u in visited:
        continue
    visited.add(u)
for (v, w) in graph.get(u, []):
    if dist[v] > dt w:
        dist[v] = dt w
        heapq.heappush(pq, (dist[v], v))
    return dist
```

3) Convert the flow chart for shortest path in an unweighted graph into code

Sol

```
from collections import deque
def shortest_path_unweighted(graph, source):
    dist = {node :- for node in graph}
        dist[source] = 0
        queue = deque([source])
    while queue:
        u = queue.popleft()
    for v in graph.get(u, [])
    if dist[v] = -1:
        dist[v] = dist[u] + 1
        queue.append(v)
    return dist
```

Convert the flowchart

code:

```
import heapq
def prim_mst(graph, start):
    visited = set([start])
    mst = []
    pq = []
    for u, w in graph.get(start, []):
        heapq.heappush(pq, (w, start, v))
    total_cost = 0
    while pq and len(visited) < len(graph)
        return mst, total cost
```

Convert the flowchart for kruskal's algorithm into code

```
class DSV:
    def __init__(self, n):
        self.parent = list(range(n))
    def find(self, x):
        if self.parent[x] != x:
            self.parent[x] = self.find(self.parent[x])
        return self.parent[x]
    def union(self, x, y):
        xx, xy, = self.find(x), self.find(y)
        if xx == xy:
            return false
        self.parent[xy] = xx
        return true
def kruskal_mst(vertices, edges):
    edges.sort(key = lambda x: x[2])
    dsv = DSV(vertices)
    mst = []
```

```
for u,v,w in edges:
        if dsv.union(u,v):
            mst.append((u,v,w))
            total_cost += w
            if len(mst) == vertices-1:
                break
        return mst, total_cost
```